

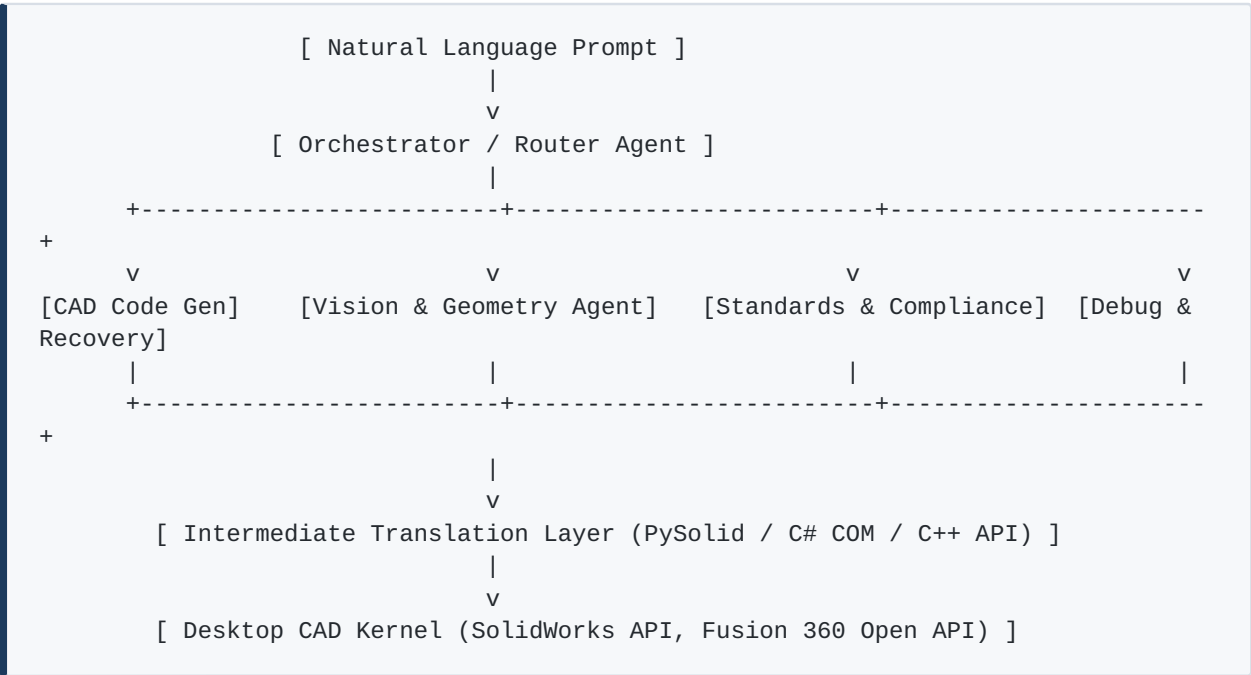
AI CAD Copilots: Architectural & Technical Specifications

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Technical Reference Document — Multi-Agent CAD Automation Systems

1. Backend Architecture & Multi-Agent System

Rather than relying on a single AI model, modern AI CAD backends use a **Multi-Agent Architecture** orchestrated via framework routing:



- **Orchestrator Agent:** Parses natural language intent, extracts dimensional constraints (e.g., “5mm fillet”, “8-bolt pattern”), and assigns sub-tasks to specialized domain agents.
- **CAD Code Generator Agent:** Translates text into programmatic CAD API code (e.g., C# / C++ COM calls for SolidWorks, Python for Fusion 360) rather than generating dumb 3D meshes.
- **Vision & Geometric Analysis Agent:** Captures spatial contexts, model B-Rep (Boundary Representation) data, topology, and active selection faces/edges directly from the CAD workspace.

- **Standards & PDM/PLM Agent:** Queries internal databases to pull exact fastener toolboxes, material properties (e.g., 6061-T6 Aluminum), title blocks, and DFM (Design for Manufacturability) guidelines.
- **Debug & Error Resolution Agent:** Monitors the CAD rebuild engine. If a feature breaks or creates a mate error, it captures the CAD system error log and dynamically attempts code healing or rollback.

2. Integration & Plugin Mechanism

- **Deployment Method:** Lightweight desktop extension DLLs compiled for Windows (e.g., SolidWorks C# Add-In, Autodesk Fusion Add-In).
- **Local vs. Cloud Execution:**
 - **Local Host Side:** The plugin hooks directly into the host CAD software's C++/C# API (SolidWorks API, Inventor API, Siemens NX Open API). It extracts the Active Document context, feature trees, geometry IDs, and sketch constraints.
 - **Cloud Backend Side:** Prompts and serialized model tree metadata are securely transmitted via WebSocket / HTTPS to cloud LLMs (e.g., GPT-4o, Claude 3.5 Sonnet, specialized fine-tuned CAD models).
- **Execution:** Cloud agents return structured API payloads, which the local plugin executes inside the CAD engine.

3. Model Latency, Time Specs, and Token Limits

Token Limits & Context Windows

- **Input Context Window:** Typically **128k to 200k tokens**. Required to pass large feature-tree histories, full BOM tables, JSON geometry trees, and company drafting rules.
- **Output Token Limit:** **4,096 to 8,192 tokens per request** (sufficient for returning procedural C#/Python scripts that modify 10 to 50 CAD operations at once).
- **Usage Allocation / Credit Limits:** Tiered credit models govern requests (e.g., Student: ~3 Million token credits/month; Professional/Enterprise: Tiered high-speed API token tiers).

Execution Time Specifications

- **Simple Feature Creation** (*Fillets, Cuts, Hole Patterns*): ~1.5 to 3 seconds.
- **Complex Multi-Step Operations** (e.g., *"Shell this part, add 4 mounting bosses, and pattern"*): ~4 to 8 seconds.
- **Full 2D Drawing & Dimension Generation:** ~10 to 25 seconds.
- **Error Diagnosis & Rebuild Fixes:** ~3 to 6 seconds per iteration loop.

4. Key Specifications Summary

Specification	Technical Details
Supported CAD Platforms	SOLIDWORKS (2020–2026), Autodesk Fusion, Inventor, Creo, NX
Plugin Architecture	Windows C# / .NET COM Add-In, Native CAD SDK Hook
Output Type	Fully editable parametric feature tree (Native <code>.SLDPRT</code> , <code>.IPT</code> , <code>.F3D</code>)
AI Backbone	Hybrid Multi-Agent LLMs + Vision Transformers + CAD Code Synthesizer
Context Window	Up to 128K - 200K Tokens per prompt call
Average Response Time	1.5 - 5 seconds per modeling command
Enterprise Connections	PDM/PLM, Local Fastener Libraries, Custom Company Drawing Standards